

A Comparative Study of Principle Component Analysis and Non-negative Matrix Factorization for Analysis of High-Density Surface Electromyography in Three Degrees of Freedom of the Index Finger

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Abstract—This study investigated the motor control mechanisms underlying index finger movements using High-Density Surface Electromyography (HD-sEMG) and force measurements. The experimental protocol involved isometric contractions in three degrees of freedom: index digit abduction, flexion, and extension. Principal Component Analysis (PCA) and Non-negative Matrix Factorization (NMF) were used to analyze the EMG data, and assess muscle activation patterns. Variance Accounted For (VAF) was calculated to evaluate the performance of PCA and NMF. This study provides insights into motor control during index finger movements and demonstrates the usability of PCA and NMF for analyzing EMG data and muscle synergies.

I. INTRODUCTION

In this study, we investigate the control mechanisms of the motor cortex in executing movements and analyze specific features of High-Density Surface-EMG (HD-sEMG) signals and motor unit behavior. The focus of our analysis is on the index finger, specifically the first dorsal interosseus muscle responsible for the movement. Understanding the intricate control of these movements is essential for gaining insights into motor control mechanisms and developing advanced rehabilitation techniques. By employing Non-negative Matrix Factorization (NMF) and Principal Component Analysis (PCA) on the recorded EMG data [1], we aim to explore the underlying patterns and synergies involved in the motor control of these finger movements.

II. METHODS

A. The Experimental Protocol

Our experimental protocol focused on investigating the motor control of the index finger through isometric contractions in three distinct degrees of freedom: index digit abduction, flexion, and extension. The aim was to analyze the muscle activation patterns during these specific movements. The experiment was conducted using the multichannel amplifier for bioelectrical systems - Quattrocento, developed by OT Bioelettronica in Turin, Italy. This state-of-the-art equipment provided high-quality recordings of High-Density Surface-EMG (HD-sEMG) signals with a bandwidth of 10–500 Hz, a resolution of 16 bits, and a sampling rate of 2048 Hz.

During the experimental sessions, a single female subject, 24 years old, performed the tasks while receiving force feedback ramps at different force levels. 64 channel electrode was placed on her dorsal interossei muscle. The experimental setup allowed us to capture and analyze the muscle activity of the muscle responsible for index finger abduction and flexion.

B. EMG Data Analysis

In order to assure proper analysis, the force data underwent a number of preprocessing processes. A two-step preparation procedure was applied on the EMG data. The dataset's outlier channels which presented with noise, artifacts, or were flat were identified and eliminated, producing a clearer EMG signal. All EMG channels were then subjected to notch filtering to remove undesirable frequency components, such as interference from power lines.

Muscle synergies are brain coordinative structures that function to reduce the computational load involved with movement and postural regulation [4]. Principal Component Analysis (PCA) and Non-negative Matrix Factorization (NMF) were performed to gain insights into the EMG data. PCA identified dominant patterns (synergies) in the EMG signals, reducing the dimensionality while retaining essential information. NMF aimed to decompose the EMG data into non-negative components, providing an alternative perspective on muscle activation patterns. All in all, we could compare the reconstructed EMG signal from both types of analysis.

The Variance Accounted For (VAF) metric was calculated for both PCA and NMF analyses to examine the performance of the dimensionality reduction strategies. The VAF quantified the proportion of variance explained by the various components or factors, and served as an evaluation criterion for the approaches' efficacy in collecting critical EMG properties. VAF is calculated the following way [2]:

$$VAF = (1 - (SSE/SST)) \quad (1)$$

where SSE is the sum of squared errors and SST the total sum of square. Note that VAF is also strongly influenced by noise level, unstructured EMG, and data filtering. Thus, it may be

difficult to compare goodness-of-fit and VAF measures across studies. [5]

By employing PCA and NMF analyses with VAF calculations, this study aimed to unravel the intricate characteristics of EMG data in the context of motor control and coordination. Comparison of reconstructed data revealed interesting results regarding the usefulness of PCA and NMF in HD-sEMG data analysis.

III. RESULTS

A. Comparison of PCA and NMF Reconstructions

Figures 1 ,2 and 3 represents the reconstructed data of the original abduction, extension and flexion EMG data by PCA and NMF analysis respectively. Only 10 channels were taken into consideration for better visualisation purposes.

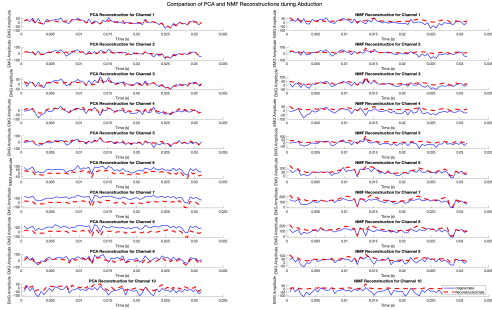


Fig. 1: Comparison of PCA and NMF Reconstruction from Abduction Dataset

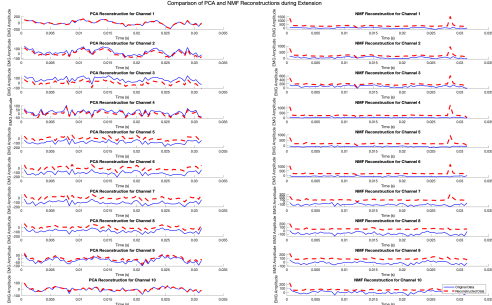


Fig. 2: Comparison of PCA and NMF Reconstruction from Extension Dataset

B. Muscle Synergies - VAF vs Number of Synergies

The results of the Variance Accounted For (VAF) vs Number of synergies is represented in the figures 4 ,5 and 6 below for abduction, extension and flexion datasets respectively. From all the figures it is apparent that VAF is significantly higher for PCA analysis than in NMF analysis.

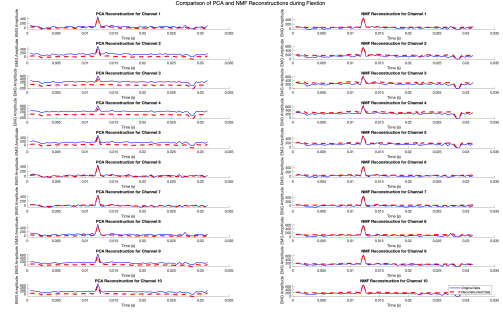


Fig. 3: Comparison of PCA and NMF Reconstruction from Flexion Dataset

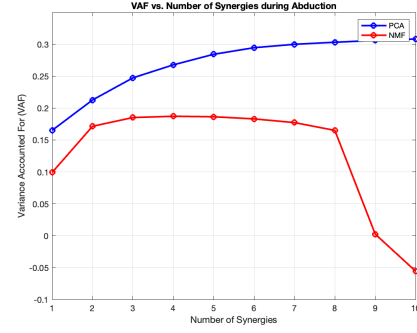


Fig. 4: VAF vs Number of Synergies from Abduction Dataset

IV. DISCUSSION

In this paper, we compared two commonly employed types of dimensional reduction techniques, Principle Component Analysis (PCA) and Non-negative Matrix Factorization (NMF) on 3 different Degrees of Freedom EMG datasets. The core nature of PCA is based on the utilization of a reduced set of principle components; by concurring this aspect, we aimed to reconstruct the original EMG data while keeping the important information. The concept of selecting the appropriate number of principle components is essential, because it determines the level of information kept in the reconstruction process. Therefore, if a sufficient number of components are retained, PCA is able to produce high-quality reconstruction.

However, in case of NMF, the original EMG data is decomposed into non-negative components and reconstructed by combining these components. The quality of reconstruction depends on achieving the balance between the number of components chosen and the sparsity.

From figures 1 and 2 it is clearly noticeable that the results provided by PCA analysis showed far more fidelity to the original abduction and extension EMG data, than the results of NMF. This could be explained by several pivotal factors. One of them is that PCA can capture significant sources of variance in linear fashion [3], resulting into effective noise filtering and keeping the most silent patterns. Opposed to PCA, NMF is challenged by selecting the adequate component number and

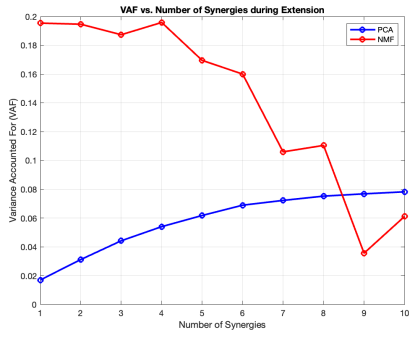


Fig. 5: VAF vs Number of Synergies from Extension Dataset

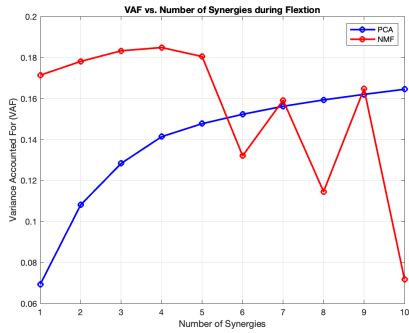


Fig. 6: VAF vs Number of Synergies from Flexion Dataset

level of sparsity. Consequently, this can result into the lower quality of data reconstruction. On the other hand, it has to be noted that figure 3 that represents reconstruction of flexion data NMF shows slightly better results than PCA analysis, but the difference is barely visible.

Apart from data reconstruction ability, in our study we decided to assess the performance of PCA and NMF in terms of Variance Accounted for (VAF) for acquiring the number of synergies from EMG data. VAF can be used as an evaluation metric for the PCA and NMF efficacy in capturing important EMG properties.

The results from figures 4, 5 and 6 indicated that in the aspect of VAF, PCA now dominated over NMF in only DoFs. Only 10 synergies were taken into consideration and from the figures it is apparent that PCA is increasing uniformly in all 3 DoFs, however, we have a sudden rapid decrease in NMF for synergies numbered 9 and 10 in abduction dataset. More rapid and inconsistent characteristics of NMF are visible in extension and flexion comparison graphs. The core idea of VAF is that the higher the VAF amount, the better. The superiority of PCA suggest that it effectively captures the important patterns of muscle activation from our EMG data, this can again be explained by the linear nature of PCA analysis. Also, PCA's ability to adjust the number of principle components in a flexible manner is a high contributor to its higher VAF performance.

V. CONCLUSION

To conclude, we compared Principle Component Analysis (PCA) and Non-negative Matrix Factorization in terms of EMG data analysis for three degrees of freedom of index finger. PCA showed superior results both in data reconstruction and VAF. Our findings highlight the effectiveness and efficiency of PCA analysis in terms of data reconstruction and capturing EMG patterns, in our case synergies. It has to be noted that we were limited in terms of number of subjects, as we only used data from 1 person, for future analysis it would be efficient to compare more extensive data. Overall, further investigations are warranted to explore the relationships between PCA and NMF in terms of VAF and data reconstruction analysis, in order to gain further insights into the advantages and disadvantages of these particular techniques in EMG data analysis.

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